OddBall - Customer Support Report (Take Home Assessment)

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Overview

This pipeline will need to compile data from a January 2025 INITIAL extract and subsequent delta files for February and March 2025 (please note that March as the boundary month is considered inclusive and interactions through the end of March 2025 SHOULD NOT be excluded). The files will be processed chronologically to arrive at the final state of records, which will be aggregated into a summary report of contact center interactions.

Data Sources:

- **Fact Table:** `interactions.csv`
- **Dimension Tables:** `agents.csv`, `contact_centers.csv`, `service_categories.csv`

Pipeline Steps:

1. Load the January initial files
2. Apply the February delta files
3. Apply the March delta files
4. Save the final state of all tables
5. Build and save the aggregated Support Report

Setup

Helper Functions

`apply_delta()`

REMEMBER: The delta files share the EXACT SAME FORMAT as the initial files, but include an action item that indicates whether a record has changed. NULLs or blanks should not be there, as it appears all of the row instances have one of the following definition changes:

Add -> Inserts a new record as a row instance. Update -> Replaces the existing record/instance (matched by a PRIMARY KEY). Delete -> Removes the record entirely (DOUBLE CHECK DELETIONS, THEY CANNOT BE UNDONE!)

`read_delta_if_exists()`

NOTE: Not every table has a delta file for every month! Example: `contact_centers` had no changes in February. To prevent the pipeline from breaking, the following helper function will skip a table silently without issuing a warning.

Step 1: Load January Initial Files

```
## agents: 5 rows

## contact_centers: 3 rows

## service_cats: 6 rows

## interactions: 10 rows
```

Steps 2 & 3: Apply February and March Deltas

Each delta file is applied in chronological order. For each month, the pipeline will process all four tables, while skipping any table that has no delta file for that month.

```
## After 202502 delta:
## agents: 7 rows
## contact_centers: 3 rows
## service_cats: 8 rows
## interactions: 18 rows
##
## After 202503 delta:
## agents: 10 rows
## contact_centers: 3 rows
## service_cats: 9 rows
## interactions: 28 rows
```

Step 4: Save Final State of All Tables

The final state of each table is persisted as a CSV in the `/output` directory.

```
## Final tables saved to /output.
```

Final Agents:

agent_id	full_name	contact_center_id	hire_date	category_id
AGT001	Sarah Johnson	CC001	2024-01-15	TECH
AGT004	Maria Garcia	CC002	2024-04-01	BENE
AGT005	James Wilson	CC003	2024-01-20	TECH
AGT006	Emily Brown	CC003	2024-05-01	ELIG
AGT007	Robert Taylor	CC001	2024-03-15	PROV

agent_id	full_name	contact_center_id	hire_date	category_id
AGT003	David Williams	CC003	2024-02-15	CLMS
AGT008	Lisa Anderson	CC002	2024-02-01	SCHD
AGT009	John Martinez	CC003	2024-04-15	GENQ
AGT010	Karen Lee	CC001	2024-06-01	DOCS
AGT002	Michael Chen	CC001	2024-03-01	TECH

Final Contact Centers:

contact_center_id	contact_center_name	location
CC001	Boston MA NE	Boston MA
CC002	Atlanta GA SE	Atlanta GA
CC003	Richmond VA E	Richmond VA

Final Service Categories:

category_id	category_name	department
TECH	Technical Support	IT
BILL	Billing Inquiry	Finance
BENE	Benefits Questions	Benefits
CLMS	Claims Processing	Operations
SCHD	Scheduling	Operations
DOCS	Documentation	Finance
GENQ	General Inquiry	Member Services
ELIG	Eligibility Check	Operations
PROV	Provider Relations	Benefits

Step 5: Build the Support Report

- **Timezone Conversion:** As noted on the instructions, all time stamps in the source data are UTC. Since the contact centers operate in Eastern Standard Time (EST), time stamps are converted to `America/New_York` before extracting the month. This will ensure interactions that occurred just after midnight UTC are attributed to the correct local date and month.
- **Department:** The `department` field is not on the interactions table directly. It is derived by joining `service_categories` via `category_id`.
- **Deleted Dimension Records:** If a contact center or service category was deleted in a delta file, interactions referencing it will produce `NA` after the join. These are replaced with `"Unknown"` per the challenge requirements, as it is unknown to us which department provided support to the caller.

```
## support_report saved to /output/support_report.csv
```

Results

Month	Contact Center	Department	Total Interactions	Total Calls	Total Call Duration (min)
2025-01	Atlanta GA SE	Benefits	2	2	22
2025-01	Atlanta GA SE	Unknown	1	0	0
2025-01	Boston MA NE	Finance	1	0	0
2025-01	Boston MA NE	Member Services	1	1	2
2025-01	Boston MA NE	Operations	1	1	19
2025-01	Richmond VA E	IT	1	1	13
2025-02	Atlanta GA SE	Member Services	1	1	8
2025-02	Atlanta GA SE	Operations	2	1	14
2025-02	Boston MA NE	Benefits	1	1	8
2025-02	Boston MA NE	Finance	1	0	0
2025-02	Boston MA NE	IT	1	1	17
2025-02	Boston MA NE	Member Services	1	1	2
2025-02	Boston MA NE	Operations	1	1	14
2025-02	Richmond VA E	IT	2	2	23
2025-02	Richmond VA E	Operations	1	1	15
2025-03	Atlanta GA SE	Operations	2	1	10
2025-03	Boston MA NE	Benefits	1	1	14
2025-03	Boston MA NE	Finance	1	1	5
2025-03	Boston MA NE	IT	2	2	38

Month	Contact Center	Department	Total Interactions	Total Calls	Total Call Duration (min)
2025-03	Boston MA NE	Operations	1	1	21
2025-03	Richmond VA E	IT	1	0	0
2025-03	Richmond VA E	Member Services	1	1	11
2025-03	Richmond VA E	Operations	1	0	0

Answers to Business Questions

Q1: Total Interactions by Contact Center in Q1 2025

Contact Center	Total Interactions (Q1 2025)
Boston MA NE	13
Atlanta GA SE	8
Richmond VA E	7

Q2: Which Month Had the Highest Interaction Volume?

Month	Total Interactions
2025-02	11
2025-03	10
2025-01	7

Q3: Which Contact Center Had the Longest Average Call Duration?

Contact Center	Total Calls	Total Duration (min)	Avg Call Duration (min)
Boston MA NE	11	140	12.7
Richmond VA E	5	62	12.4
Atlanta GA SE	5	54	10.8